

System Study

Reposys — Campus Reprography Automation System

1. Study of the Existing System

The reprography counter at Saintgits College of Engineering currently operates as a fully manual, walk-in service. Students and faculty who require printing, photocopying, scanning, or binding services must physically visit the counter during operating hours.

The existing workflow is as follows:

- The student arrives at the counter and waits in a physical queue
- The student verbally communicates print requirements to the counter staff
- Staff manually configures the print job based on verbal instructions
- Payment is collected in cash at the counter
- The student either waits at the counter or returns later with no notification mechanism
- Staff manages all pending jobs mentally or through informal paper notes
- The administrator reviews daily cash collections manually at end of day

There is no digital record of orders placed, no formal queue management system, no payment trail, and no mechanism for students to track the status of their orders remotely.

2. Drawbacks of the Existing System

The following problems have been identified in the current manual system:

- **No remote order placement:** Students must be physically present at the counter to place an order, resulting in loss of time during working hours
- **Unstructured queue:** There is no formal priority system. Orders are processed based on staff judgment, leading to inconsistencies
- **No order tracking:** Once an order is placed, the student has no way to know when it will be ready without physically visiting the counter again
- **Cash-only transactions:** All payments are collected in cash. This creates reconciliation difficulties, increases the risk of errors, and leaves no digital payment trail
- **No receipt or proof of payment:** Students do not receive formal receipts for transactions
- **No inventory monitoring:** Staff have no system to track paper stock, toner levels, or binding materials. Stockouts are discovered only when work cannot be completed
- **No operating hour notifications:** Students have no way to know in advance if the counter is closed for a holiday or operational reason
- **No walk-in self-service:** Users without prior knowledge of the counter must ask staff for pricing, service availability, and process information
- **No audit trail:** There is no record of who placed which order, what was charged, or who processed the job

- **No group cost sharing:** When groups of students share a print job, splitting payment is handled informally with no system support

3. Proposed System

Reposys — Campus Reprography Automation System — is a full-stack web-based management system proposed to replace the manual counter workflow at Saintgits College of Engineering.

The proposed system will allow students and faculty to place print, photocopy, scan, binding, digitally from any device without visiting the counter. Orders will enter a structured priority queue managed through a staff dashboard. Payments will be accepted through online payment gateway, internal digital wallet, or cash at counter.

Students will receive real-time status updates and push notifications when their order is ready. Staff will process orders through a digital queue interface with pickup verification through one-time password. Administrators will have a management dashboard for pricing configuration, user management, inventory tracking, and operational reporting.

The system will also support walk-in users without college accounts through a guest kiosk mode using email-based one-time password authentication.

The proposed system will:

- Allow students and faculty to place and track orders from any device at any time
- Provide staff with a structured, priority-sorted queue dashboard
- Support three payment methods: online payment, digital wallet, and cash at counter
- Send real-time notifications and push alerts for order status changes
- Allow cost splitting among students for group print jobs
- Provide administrators with full system configuration and monitoring capability
- Support walk-in guest users without requiring account registration
- Function as an installable Progressive Web App on mobile devices

4. Feasibility Study

Before committing to the development of Reposys, a feasibility study was conducted to evaluate whether the proposed system is viable from technical, operational, economic, schedule, and legal perspectives.

4.1 Technical Feasibility

Technical feasibility determines whether the required technology exists and whether the development team has the capability to implement it.

The proposed system requires:

- A responsive web interface usable on both desktop and mobile devices
- Secure user authentication with multiple role levels
- Real-time communication between the server and connected clients

- File upload, validation, and cloud-based storage
- Integration with a payment gateway
- Background task scheduling for queue management and notifications
- Push notification delivery to mobile devices

All of the above requirements can be met using existing, well-established technologies:

- React and Vite provide a mature framework for building multi-step interactive web interfaces
- JSON Web Tokens stored in secure cookies provide a proven mechanism for session-based authentication
- Socket.io is a widely deployed library for real-time bidirectional communication between server and browser
- Cloudinary is an established cloud media platform supporting file upload, storage, and access control
- Razorpay is India's most widely used payment gateway with support for UPI, cards, and net banking
- node-cron provides lightweight task scheduling within a Node.js environment
- The Web Push API with VAPID key authentication is a W3C standard for delivering push notifications through service workers

No proposed feature requires experimental technology, custom hardware, or proprietary software that is not available to the development team.

Conclusion: Technically Feasible

4.2 Operational Feasibility

Operational feasibility determines whether the proposed system will work within the real environment it is deployed in and whether the intended users will be able to use it effectively.

Counter staff:

Counter staff are the primary operational users of the system. They are not expected to have technical backgrounds. The proposed staff interface must present a simple, action-oriented queue view that requires minimal training. The system must not add complexity to the physical counter workflow but must instead reduce the administrative burden currently placed on staff.

Students and Faculty:

Students and faculty will use the system primarily from mobile devices. The order placement flow must be simple, fast, and completable without prior training. Since many users will access the system from mobile browsers, the interface must be responsive and the system should be installable as a Progressive Web App to provide a native app-like experience.

Walk-In Users:

Not all users who require reprography services will have a college account. The proposed guest kiosk mode addresses this by allowing walk-in users to authenticate using only an email address and a one-time password, without requiring account registration.

Administrator:

The system administrator must be able to configure pricing, manage users, and review system

performance without technical intervention. All configuration must be accessible through a web-based management dashboard.

Connectivity Risks:

Internet connectivity disruptions at the counter are a realistic operational risk. The system must handle this gracefully by providing an offline fallback page and ensuring that orders already submitted remain safe in the database regardless of subsequent connectivity issues.

Conclusion: Operationally Feasible

4.3 Economic Feasibility

Economic feasibility determines whether the financial cost of developing and operating the proposed system is justified by the benefits it will provide.

Estimated Infrastructure Costs:

Service	Purpose	Estimated Monthly Cost
MongoDB Atlas (Mo Free Tier)	Database hosting	₹0
Cloudinary (Free Tier)	File storage	₹0
Vercel (Hobby Free Tier)	Frontend hosting	₹0
Oracle Cloud Always Free (Ampere A1)	Backend hosting	₹0 (permanently free)
Razorpay	Payment gateway	Transaction fee only (no monthly charge)

The total fixed infrastructure cost for an academic deployment is zero. All required services offer free tiers sufficient for the expected usage volume of a single college reprography counter.

Estimated Benefits:

- Reduction in staff time spent on verbal order taking, cash handling, and manual queue management
- Reduction in student time lost to physical queuing
- Elimination of cash reconciliation errors through digital payment records
- Prevention of inventory stockouts through proactive stock monitoring and alerts
- Elimination of wrong-person pickup incidents through OTP-based pickup verification
- Creation of a full audit trail for all orders, payments, and administrative actions

Conclusion: Economically Feasible

4.4 Schedule Feasibility

Schedule feasibility determines whether the proposed system can be designed, developed, and tested within the available project timeline.

The development of Reposys is planned in sequential phases:

Phase	Scope
Phase 0	Project setup, infrastructure configuration
Phase 1	User authentication and role management
Phase 2	Document upload, validation, and cloud storage
Phase 3	Order wizard, order management, payment integration
Phase 4	Real-time queue infrastructure
Phase 5	Admin dashboard and system configuration
Phase 6	Notifications, complaint system, reporting
Phase 7	Guest kiosk mode and PWA implementation

Each phase produces a working, testable increment. If any phase experiences delay, all prior phases remain functional and demonstrable independently.

The primary schedule risk is integration with multiple third-party APIs (Razorpay, Cloudinary, CloudConvert, Gemini API), which may behave differently from their documentation in edge cases. This risk will be mitigated by using mock implementations during early development and replacing them with live integrations during integration testing.

Conclusion: Schedule Feasible

4.5 Legal and Ethical Feasibility

Legal and ethical feasibility determines whether the proposed system raises any concerns related to data privacy, security, or ethical use of technology.

- Student names, email addresses, and college IDs will be stored as part of the user account. This is standard institutional data appropriate for a campus management system.
- Payment card and UPI data will not be stored on Reposys servers at any point. Razorpay handles all sensitive payment information. Reposys will store only payment reference identifiers and verified payment status.
- User passwords will be stored as one-way cryptographic hashes. Plaintext password storage will not be permitted.
- Authentication tokens will be stored in httpOnly cookies, preventing access by JavaScript and reducing the risk of cross-site scripting attacks.
- Documents uploaded by students for printing are personal academic files. Access will be restricted to the document owner, assigned counter staff, and system administrator.

- All administrative actions will be recorded in an immutable audit log. Log entries cannot be modified or deleted after creation.
- Guest user sessions will be time-limited and will not retain personal data beyond the active session.

Conclusion: Legally and Ethically Feasible

5. Modules of the Proposed System

Based on the system study and feasibility analysis, the proposed Reposys system is organized into the following modules:

Module 1: User Authentication and Access Control

This module manages all aspects of user identity and system access. It covers registration, email verification, login, session management, and role-based access enforcement across all parts of the system.

Features:

- User registration for students and faculty
- Email verification before account activation
- Faculty role approval by administrator
- Secure login with session token management
- Role-based access control for all system areas
- Password reset via email
- Multi-device session tracking and remote logout
- Administrator account provisioned through system seed, not self-registration

Module 2: Document Management

This module handles the intake, validation, analysis, and secure storage of files submitted for printing or other services.

Features:

- File upload with type and content validation
- Automated document analysis (blank page detection, image quality check, page count, password protection detection)
- Support for uploading multiple documents in a single order
- Server-side conversion of Word documents to PDF format
- Document preview before order submission
- Cloud-based file storage with access-controlled retrieval

- Free document utility tools (image resize, image compress, PDF resize, format conversion) independent of the order workflow

Module 3: Order Management

This module manages the full lifecycle of a reprography order from placement through to completion.

Features:

- Multi-step order placement wizard
- Support for service types: Printing, Photocopying, Scanning, Binding, Document Conversion
- Print configuration: copies, color mode, page sides, paper size, binding type, special instructions
- Server-side cost estimation based on administrator-configured pricing
- Unique order token generation per day per service prefix
- Order priority scoring based on user role and preferred pickup time
- Complete order status history with timestamps and actor records
- Order cancellation from queue
- Partial order completion handling
- Order repeat from previous completed orders
- Preferred pickup time slot selection

Module 4: Queue Management

This module manages the physical counter processing queue in real time.

Features:

- Separate priority queues per service type
- Priority-based ordering with aging mechanism to prevent indefinite waiting
- Staff order locking to prevent concurrent processing conflicts
- Real-time queue updates to staff dashboards
- Estimated wait time calculation per order

Module 5: Payment Management

This module handles all financial transactions within the system.

Features:

- Online payment via Razorpay (UPI, cards, net banking)
- Internal digital wallet with top-up via Razorpay
- Cash payment with collection at counter during pickup

- Atomic payment processing to prevent inconsistent financial states
- Full transaction history per user
- Refund processing for cancelled paid orders
- Receipt generation for completed orders

Module 6: Notifications and Communication

This module ensures all users receive timely information about orders, payments, and system events.

Features:

- Real-time in-browser order and queue status updates
- Persistent in-app notification history
- Push notifications delivered to mobile devices even when the browser is closed
- Email notifications for order ready alerts, OTP delivery, and account events
- Order-linked complaint and messaging system between students and staff
- AI-powered chatbot assistant for student queries
- Direct and group messaging between students and faculty
- Order rating and feedback after completion

Module 7: Group Orders and Split Payment

This module allows multiple students to share the cost of a single print order.

Features:

- Group order creation during the payment step of the order wizard
- Per-participant cost assignment by the order creator
- Live split payment card visible in friend chat
- Real-time payment status updates per participant
- Atomic wallet transactions for split payment settlement

Module 8: Administration and System Management

This module provides administrators and staff with the tools to configure, monitor, and operate the system.

Features:

- Administrator dashboard with real-time system statistics
- Pricing and order limit configuration
- User account management and role approval

- Immutable system activity audit log
- Shop operating hours management with automatic schedule mode
- Inventory stock monitoring with low-stock alerts and usage forecasting
- Operational reports and analytics
- Staff operational report submission and acknowledgement

Module 9: Guest Kiosk Mode

This module provides a walk-in order placement capability for users without college accounts.

Features:

- Email-based one-time password authentication
- Time-limited guest session without account registration
- Full order placement and document upload within the session
- Cash-only payment with counter collection
- Public order tracking page accessible via QR code without login
- Automatic session expiry on order completion or inactivity

Module 10: Progressive Web App

This module enables Reposys to be installed on mobile devices and to deliver notifications and offline functionality beyond the browser tab.

Features:

- Home screen installation on Android and iOS devices
- Offline fallback page when network is unavailable
- Background push notification delivery via service worker
- Standalone display mode without browser navigation elements

6. Summary of Feasibility

Feasibility Type	Conclusion
Technical	Feasible — all required technologies are mature and available
Operational	Feasible — system fits the working environment of all user roles
Economic	Feasible — near-zero infrastructure cost for academic deployment
Schedule	Feasible — phased development allows incremental delivery
Legal and Ethical	Feasible — no data privacy or ethical concerns that cannot be mitigated